8-bit 10 MS/s SAR ADC in SkyWater 130 nm

Design, simulation, layout and physical verification with an open-source flow

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Scope. This is a design-and-verification project. The ADC was designed, simulated at schematic level and after parasitic extraction, laid out, and verified with DRC and LVS. It was not fabricated; every result below comes from simulation or from the verification tools.

Summary

Parameter	Target	Schematic (TT, 27 C)	Extracted layout (TT, 27 C)
Resolution	8 bit	8 bit	8 bit
Sample rate	10 MS/s		
Input range (differential)	3.6 Vpp		
SNDR, fin = 430 kHz		47.6 dB	49.2 dB
SNDR / ENOB, fin = 4.96 MHz	>= 46.9 dB / 7.5 b	48.4 dB / 7.74 b	49.2 dB / 7.89 b
SFDR, fin = 4.96 MHz		59.5 dB	60.4 dB
|DNL| max / |INL| max	< 0.5 / 0.5 LSB	0.62 / 0.58 LSB	0.16 / 0.11 LSB
Missing codes	0	0	0
Power (VDDA + VDDD)	< 500 uW	68 + 55 uW	91 + 144 uW
Walden FoM		57 fJ/conv-step	99 fJ/conv-step
Area (incl. routing)	< 40000 um2		29994 um2 (372.3 x 80.6 um)
Worst PVT corner ENOB (4.96 MHz)			7.43 b at sf 27C 1.80V (schematic)
Comparator offset sigma (MC)	< 3 mV	3.17 mV	
DRC (Magic / KLayout) / LVS	clean		0 / 0 errors; Circuits match uniquely.

Supply and reference are ideal sources (VDDA = VDDD = VREF = 1.8 V); see Limitations.

1. Architecture

The converter is a fully-differential charge-redistribution SAR with top-plate sampling and monotonic (switch-up) capacitor switching. Because the input is sampled on the top plates, the MSB decision needs no DAC transition, and each array needs only 128 unit capacitors instead of 256: after each decision, one capacitor on the lower side switches from ground to VREF. The top-plate common mode therefore rises from 0.9 V toward VDD during a conversion, which is why the comparator uses an NMOS input pair. One conversion takes 10 cycles of a 100 MHz clock (1 sample, 8 compare, 1 output), giving 10 MS/s. The LSB is 14.06 mV.

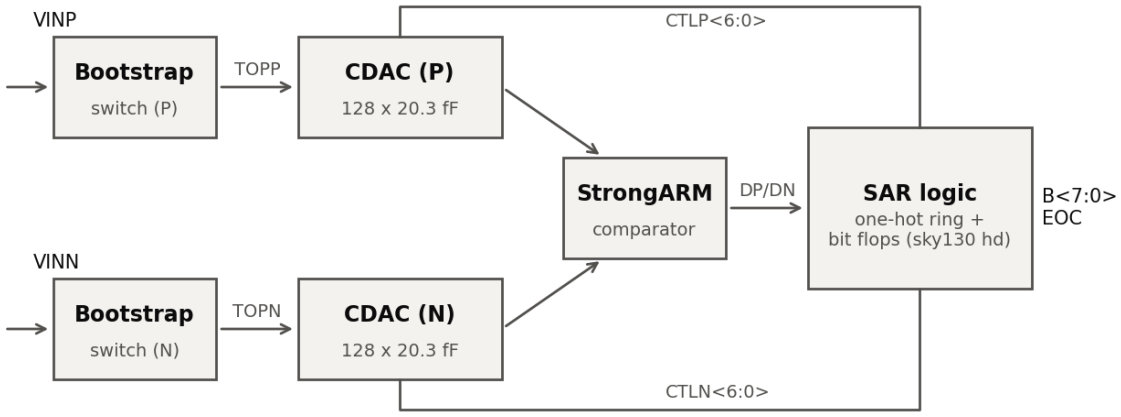


Figure 1. Block diagram. Each side has a bootstrapped sampling switch and a binary MIM-capacitor array driven by sky130_fd_sc_hd inverters; the SAR logic is built from standard cells.

Resolution trade study. The number of bits and the unit capacitor were chosen with a Monte Carlo study of the exact switching algorithm, using the PDK's MIM matching coefficient (unit sigma 0.93 % for a 3 x 3 um unit). At 8 bits, mismatch alone gives a 99th-percentile |DNL|max near 0.2 LSB, so 8 bits is not mismatch-limited, while still making comparator offset, noise and layout parasitics visible at the LSB level.

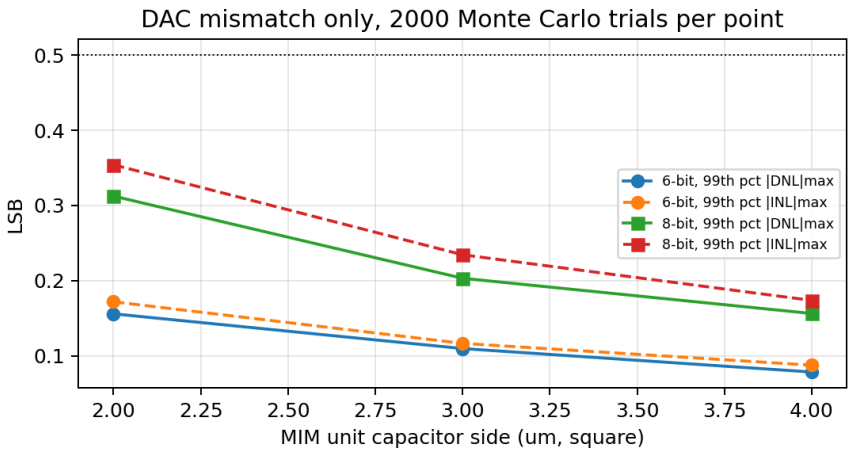


Figure 2. Stage-1 trade study: DAC-only linearity versus unit-capacitor size (2000-trial Monte Carlo, PDK MIM mismatch model).

2. Circuit design

2.1 Comparator

A StrongARM latch with an NMOS input pair (W/L = 2/0.5 um x 4), PMOS reset of both the output and the internal X nodes, and equal inverter loads on both outputs. A NOR SR latch in the logic holds each decision. At 0.9 V common mode it resolves 1 mV in 340 ps and 0.1 mV in 405 ps. Its input-referred noise is 322 uV rms. ngspice's trnoise source aborts with these models, so the noise was obtained by injecting a random source sized to the input pair's thermal noise and fitting a probit curve to the decision statistics.

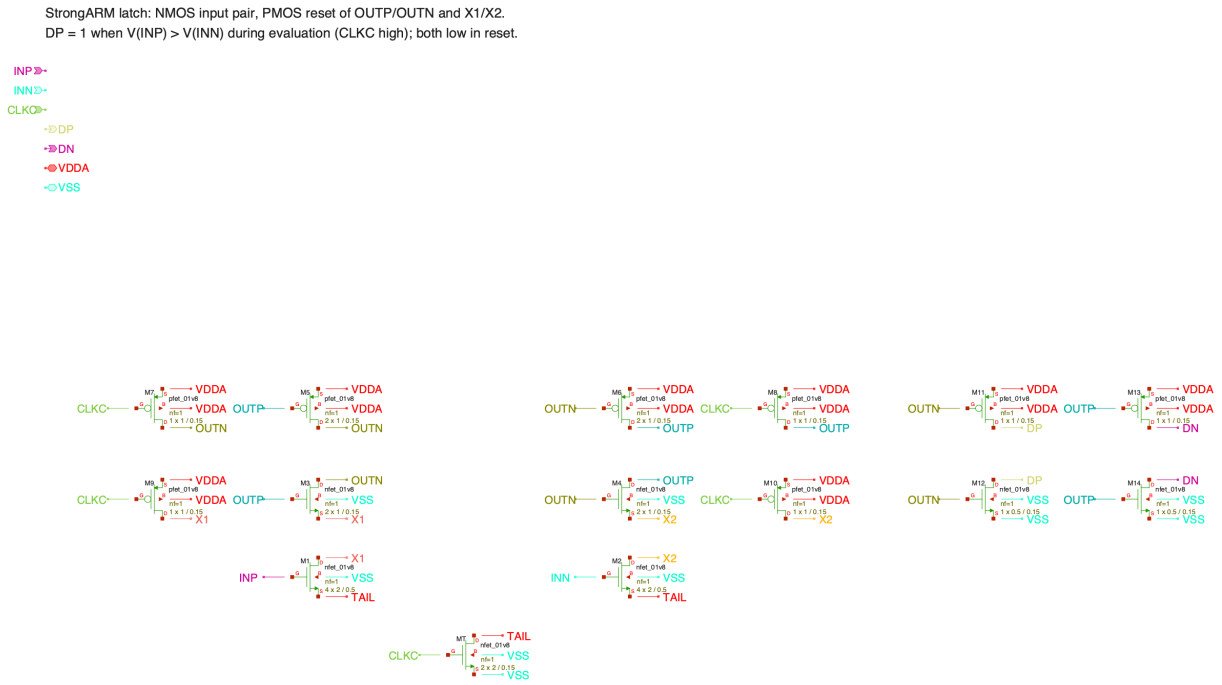


Figure 3. Comparator schematic (xschem; nets connected by label).

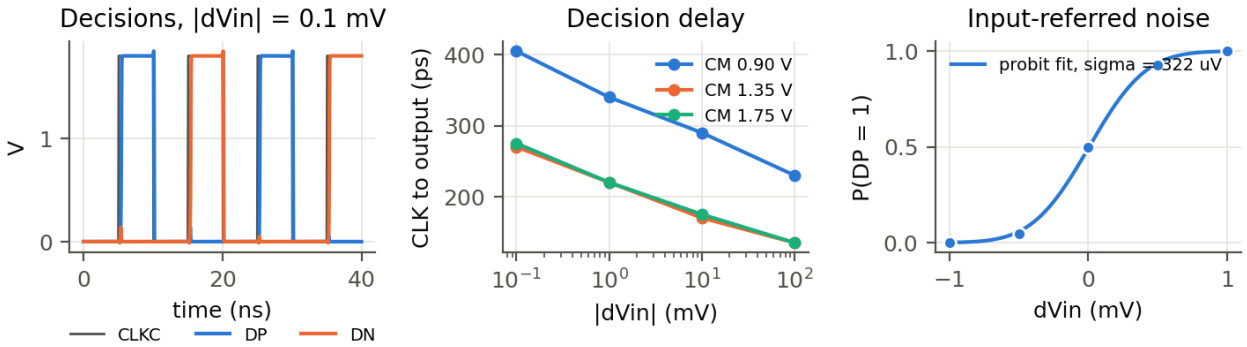


Figure 4. Comparator transient decisions, delay versus input and common mode, and the noise probit fit (schematic, TT, 27 C).

2.2 Bootstrapped sampling switch

Each input is sampled through an NMOS switch (W = 5 um x 2) whose gate is bootstrapped by a 10 x 10 um MIM capacitor, so its on-resistance stays nearly independent of the 0 to 1.8 V input.

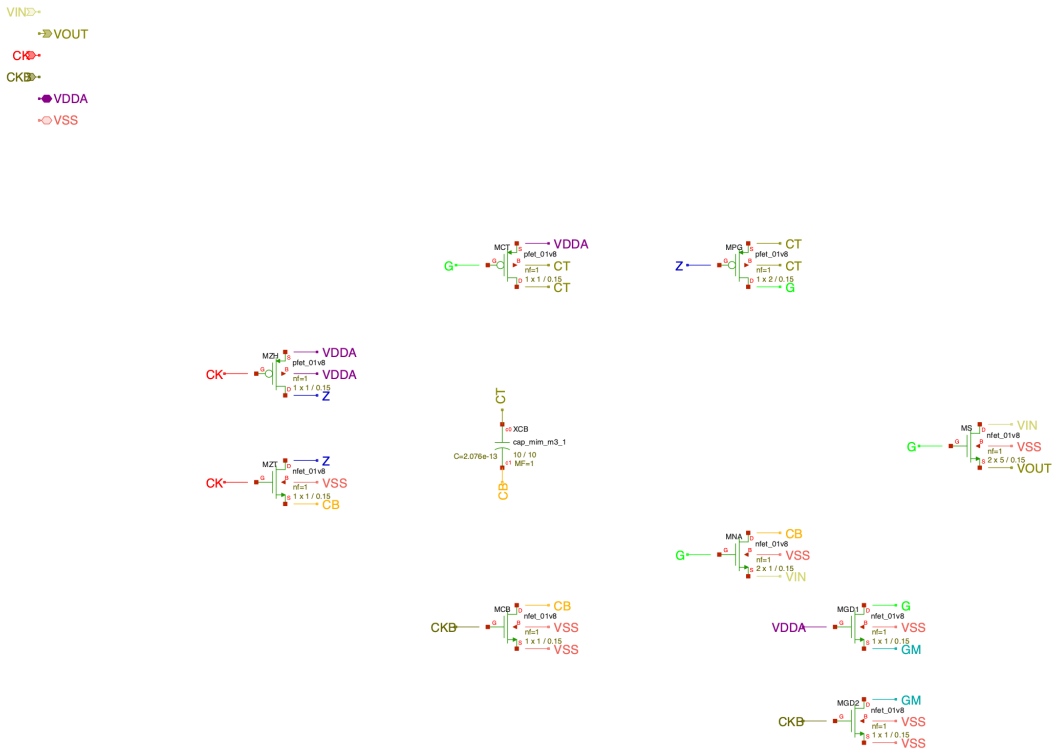


Figure 5. Bootstrapped switch schematic (xschem).

2.3 Capacitor DAC and drivers

Each side has 128 units (2.60 pF) plus 52 edge dummies. The bottom-plate drivers are sized per bit (inv_12 for the MSB group down to inv_1), because with inv_4 drivers the MSB group took 4.1 ns to settle to 0.5 LSB, too close to the 5 ns half-cycle. With the final sizing, every bit settles to 0.1 LSB within 1777 ps.

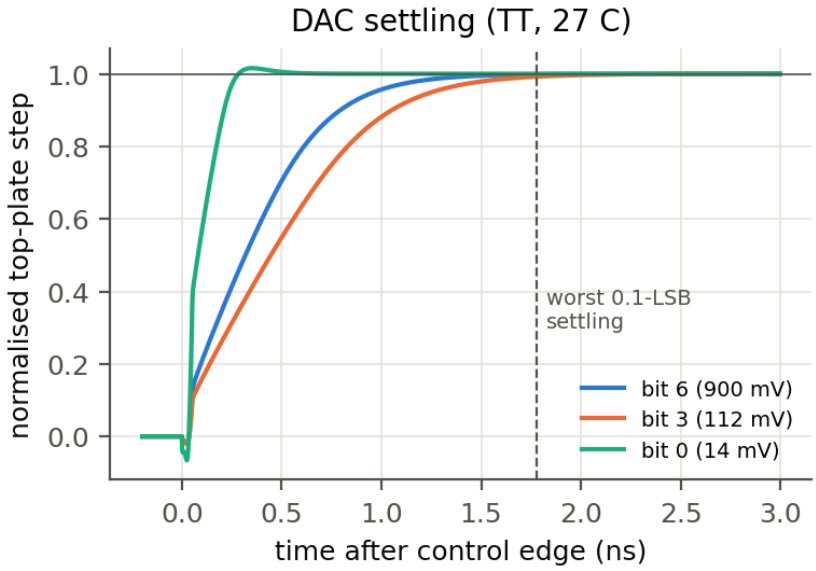


Figure 6. Top-plate settling for three bits after their driver switches (schematic).

2.4 SAR logic

A 10-flop one-hot ring sequences the conversion, and each bit is captured in a DFF clocked by the next token, with an asynchronous reset during sampling. The logic passed 576 of 576 checks over 12 forced decision sequences (0 failures).

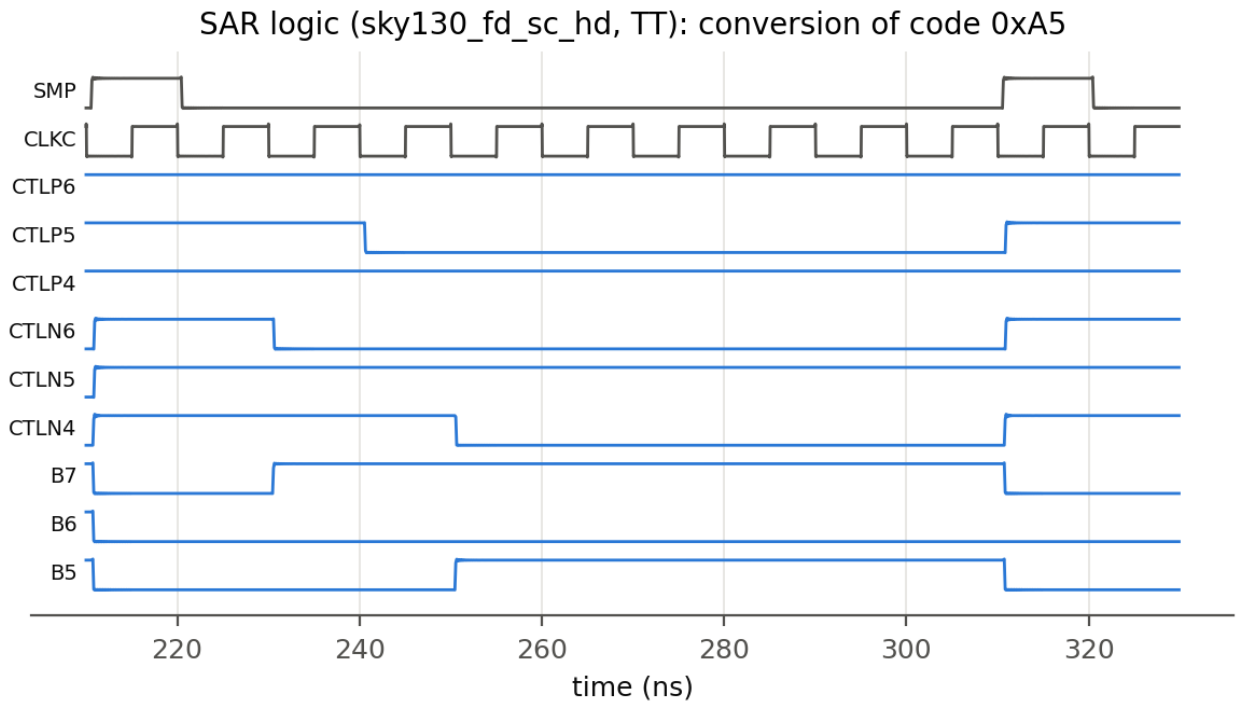


Figure 7. SAR logic waveforms for one conversion (forced comparator decisions). CTLP/CTLN are the active-low inputs of the DAC driver inverters.

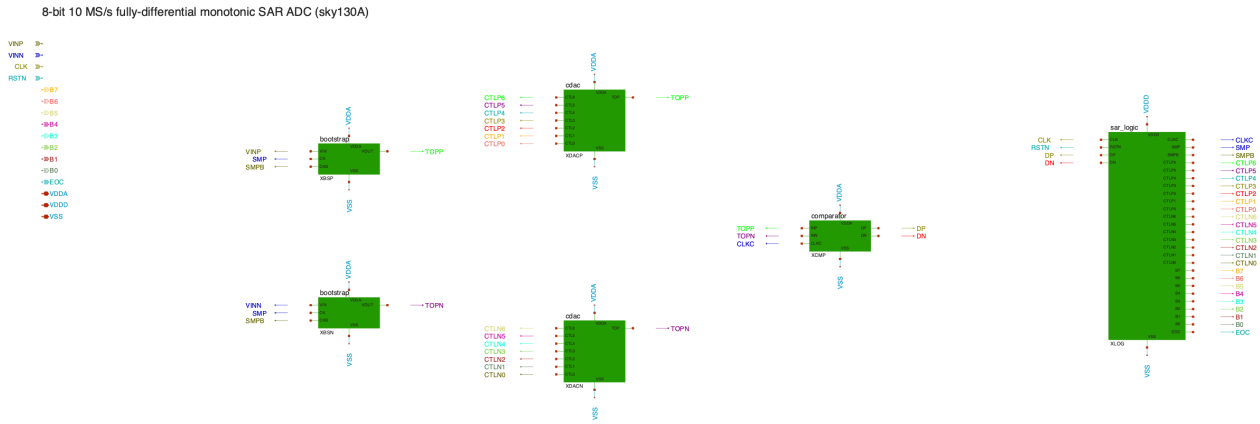


Figure 8. Top-level schematic (xschem).

3. Full-ADC simulation (schematic, TT, 27 C)

Full transistor-level runs use ngspice with the sky130 models and the sky130_fd_sc_hd SPICE netlists for the logic. Long runs are split into independent time windows that are simulated in parallel, and each window starts with one discarded conversion. Without that lead-in, the first conversion of each window was wrong because the bootstrap capacitor starts uncharged.

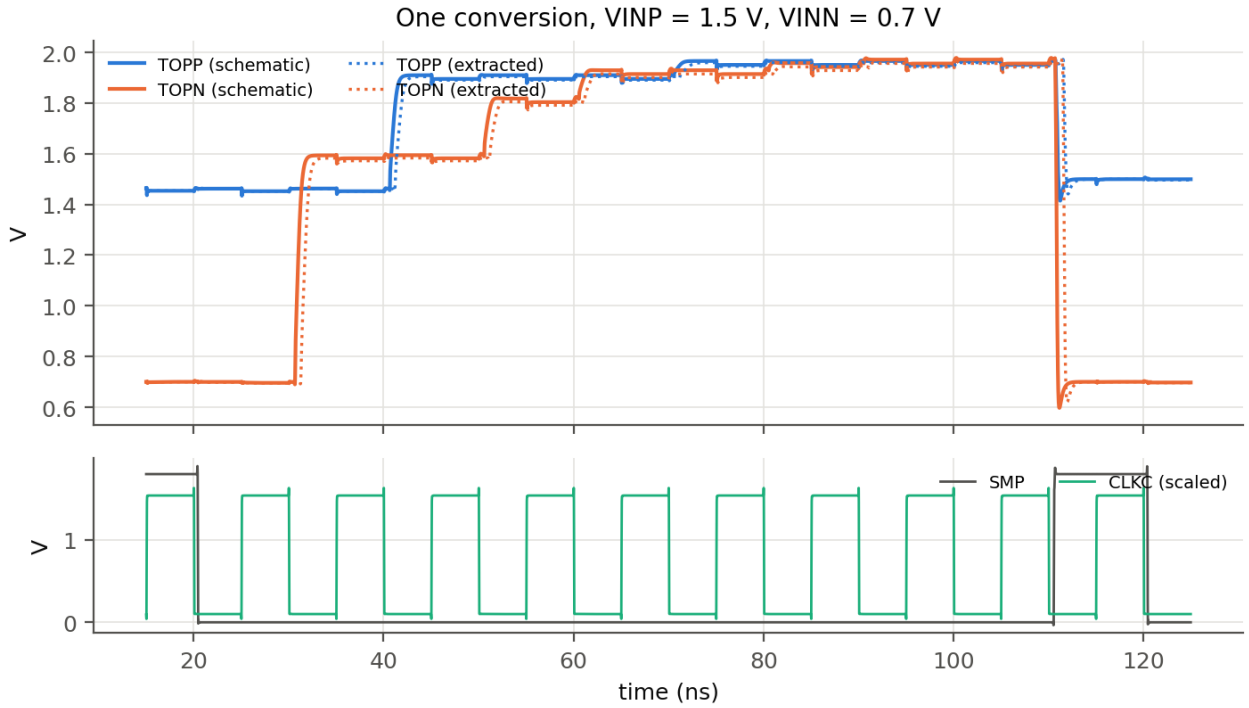


Figure 9. Internal waveforms of one conversion. The top plates are sampled and then rise monotonically as one capacitor per side switches up after each decision.

Dynamic performance uses 256-point coherent FFTs at -1 dBFS. SNDR is 47.6 dB (ENOB 7.61) at 430 kHz and 48.4 dB (ENOB 7.74) at 4.96 MHz. Static linearity comes from a 15-hits-per-code ramp: |DNL|max 0.62 LSB and |INL|max 0.58 LSB. The fitted gain error is +2.54 %, as expected: the top-plate parasitic capacitance attenuates the DAC steps but not the sampled input, so more steps are needed per volt.

Linearity misses the target at schematic level. |DNL|max 0.62 LSB and |INL|max 0.58 LSB exceed the 0.5 LSB targets. The capacitors are ideal in this netlist, so this is not mismatch. The error is concentrated at major carries: the 31/32, 63/64 and 191/192 transitions are displaced by about 0.4 LSB in opposite directions, and the ramp output toggles back and forth near transitions although the testbench has no noise source. It is not a simulator timestep artifact. Converting DC inputs around the 63/64 transition with a 10x smaller maximum step (0.01 ns) moved the transition by less than 2 mV (`scripts/diag_tstep.py`). The extracted layout does not show the effect (Section 6). The root cause was not isolated, and it is the main open item of this design.

4. PVT corners and Monte Carlo

Full ADC at 4.96 MHz (near Nyquist), -1 dBFS of the supply-scaled full scale, across process (tt/ss/ff/sf/fs), temperature (-40/27/125 C) and supply (1.62/1.8/1.98 V).

Corner	SNDR (dB)	SFDR (dB)	ENOB (b)	Power (uW)
tt 27C 1.80V	48.4	59.5	7.74	123
ss 27C 1.80V	48.4	59.9	7.75	123
ff 27C 1.80V	48.2	59.2	7.71	124
sf 27C 1.80V	46.5	54.0	7.43	125
fs 27C 1.80V	48.8	58.2	7.81	122
ss 125C 1.62V	48.2	61.0	7.72	103
ff -40C 1.98V	47.6	58.6	7.61	149
tt -40C 1.80V	48.4	58.7	7.75	120
tt 125C 1.80V	48.0	60.2	7.69	129
tt 27C 1.62V	47.7	55.8	7.63	98
tt 27C 1.98V	48.3	61.3	7.73	151

Corners below the 7.5-bit ENOB target: sf 27C 1.80V. Every other corner meets it.

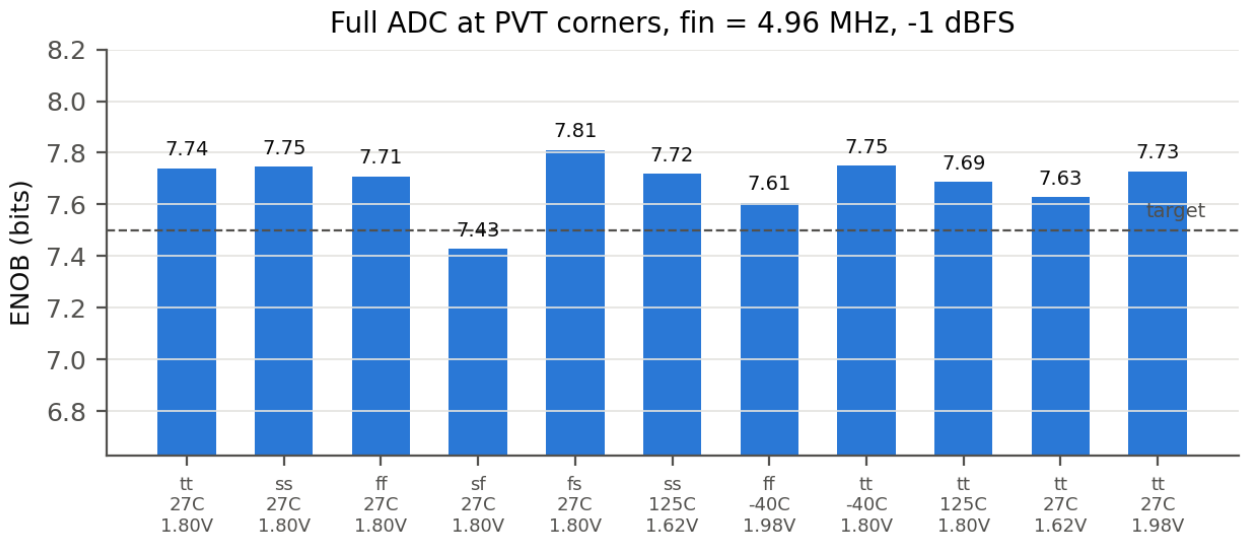


Figure 10. ENOB across PVT corners (schematic). Dashed line: target.

Mismatch. The sky130 *_mm libraries randomise transistor parameters per run. Over 200 runs at 0.9 V common mode, the comparator offset has a mean of 0.06 mV and a sigma of 3.17 mV. At 1.75 V the sigma is 10.23 mV, so the 0.9 V sigma slightly exceeds the 3 mV target. In this architecture a fixed comparator offset only shifts the transfer curve and does not change DNL or INL. These libraries do not randomise the MIM capacitors, so capacitor mismatch was simulated with the behavioural model of the same switching sequence (2000 trials, unit sigma 0.93 %): 99th-percentile |DNL|max 0.22 LSB, |INL|max 0.24 LSB, and 100.0 % of trials with both below 0.5 LSB. Full-ADC transistor mismatch runs (8 seeds, 430 kHz) give ENOB between 7.42 and 7.69 b.

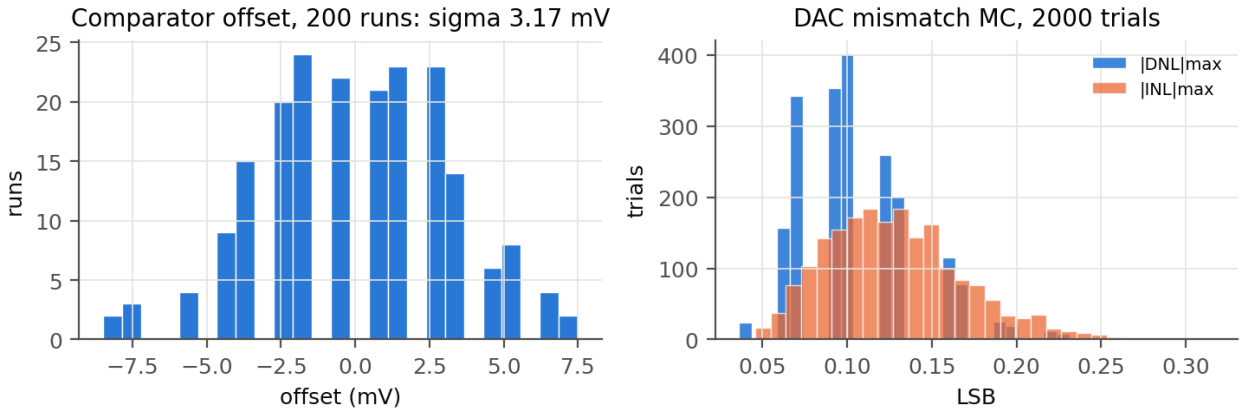


Figure 11. Left: comparator offset distribution (transistor-level Monte Carlo). Right: DAC linearity distribution from capacitor mismatch (behavioural Monte Carlo).

5. Layout and physical verification

The layout is generated by a Python script that drives Magic (parametric transistors with all terminals on metal-1 buses, PDK standard cells, MIM arrays, and a met3 channel router with met2 stubs). From left to right the floorplan is bootstrap P, DAC P, comparator, DAC N, bootstrap N, with a standard-cell row of P drivers, logic and N drivers below. The unit-capacitor pitch is 5 um (MIM spacing rule capm.11). The macro is 372.3 x 80.6 um.

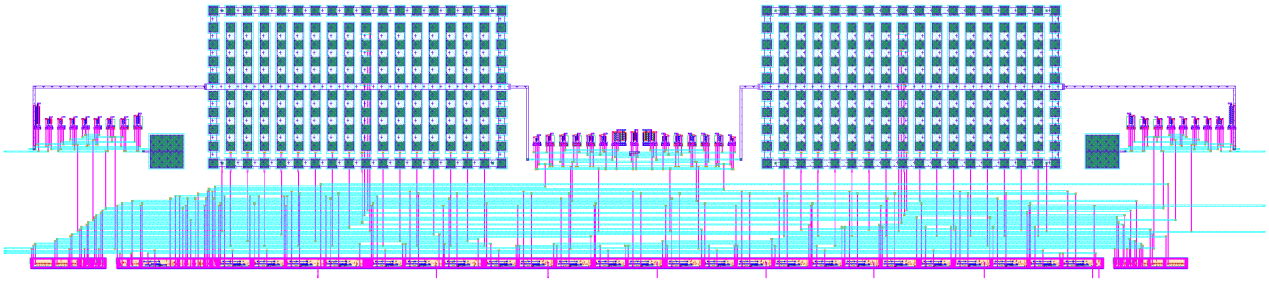


Figure 12. Full layout (KLayout render of the GDS).

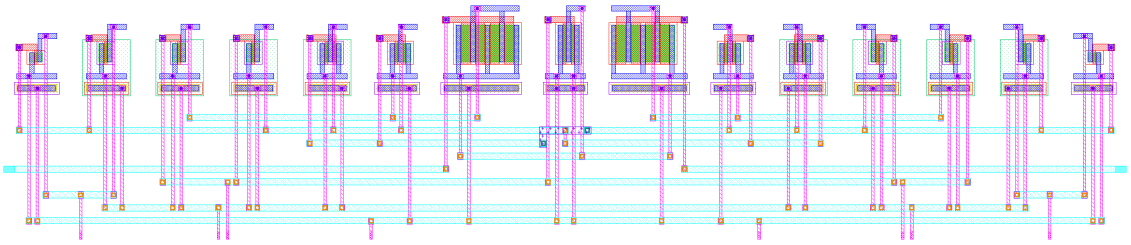


Figure 13. Comparator layout (v2): mirror-symmetric placement and routing. OUTN and OUTP swap tracks at the centre line, so each sees the same surroundings.

DRC. Magic (full sky130A deck) and KLayout (sky130A_mr, FEOL + BEOL + off-grid) both report 0 errors. The first pass had 8696 Magic error tiles. The main causes were vias drawn at cut size instead of with their enclosure, a too-tight MIM pitch, and via2 placed under capm. A canary cell with 3 deliberate violations confirms that both decks catch real errors. One DRC failure exposed a real connectivity bug: the bootstrap clock exits kept their block-local names, so the sampling switches were not connected to the logic.

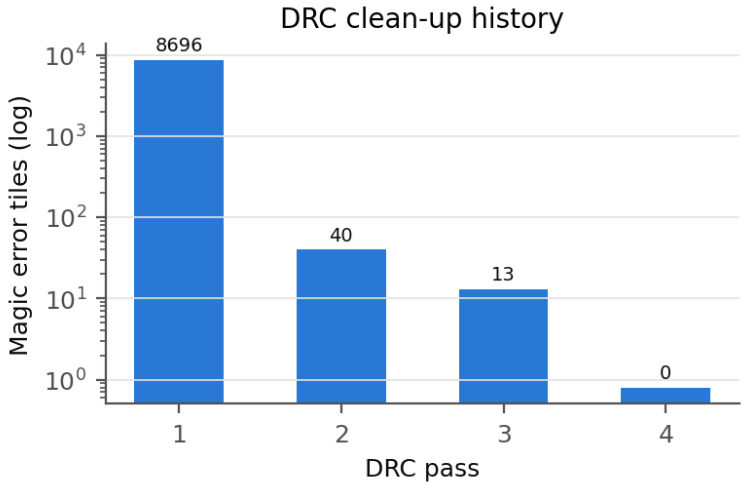


Figure 14. Magic DRC error count per clean-up pass.

LVS (netgen, sky130A setup):

Cell	Devices (layout / schematic)	Nets	Result
comparator	15 / 15	12 / 12	Circuits match uniquely.
bootstrap	10 / 10	11 / 11	Circuits match uniquely.
sar_adc	67 / 67	73 / 73	Circuits match uniquely.

Fixing LVS found a missing exit via2 and a real short between the bootstrap capacitor and CTLN6 (see signoff/lvs/lvs_history.md).

6. Post-layout simulation

Parasitics are extracted with Magic (capacitance to substrate plus coupling capacitance; no resistance). The extracted netlist is wrapped with the schematic port order, so every testbench runs unchanged on the layout.

Comparator symmetry. The first comparator layout (v1) had mirror-symmetric placement but was routed by a generic left-edge channel router. In extraction, OUTN had 13.8 fF against 11.4 fF on OUTP. That produced an offset that depends on common mode, 6.5 mV at 0.9 V and more than 60 mV at 1.75 V, and the monotonic scheme makes its last decisions near 1.75 V. All codes shifted by -7 LSB. The v2 layout mirrors the routing as well as the placement.

An extraction artifact. With v2, the extracted coupling capacitances matched to within 0.01 fF, yet the offset was still 5 to 19 mV. Two controlled re-simulations isolated the cause. Deleting every extracted capacitor left the offset unchanged, while deleting only the diffusion area and perimeter parameters (ad/as/pd/ps) brought it below 1.3 mV. Hierarchical extraction had counted 1.16 um2 of one internal node's diffusion twice, at the boundary between normal and mirrored sub-cells. Flattening the cell before extraction fixed it, so the remaining 5 to 19 mV was a tool artifact, not a circuit effect.

With v2 and flat extraction, the comparator offset is 0.75 / 1.25 / 1.75 / 1.25 mV at 0.9 / 1.2 / 1.5 / 1.75 V common mode (schematic: 0.25 mV; the sweep step is 0.5 mV). The full ADC returns the same codes as the schematic at all five DC test inputs.

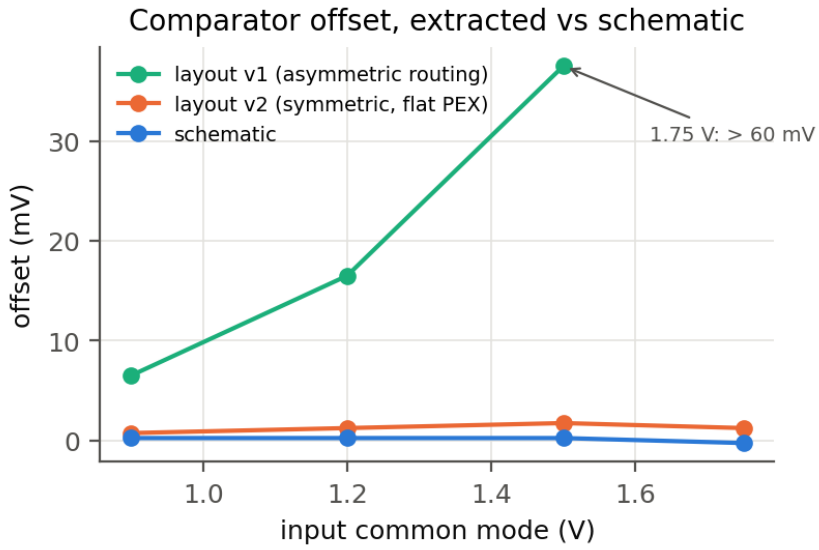


Figure 15. Comparator offset versus input common mode: schematic, extracted v1 and extracted v2.

Dynamic and static, extracted layout. SNDR is 49.2 dB at 430 kHz and 49.2 dB (ENOB 7.89) at 4.96 MHz. The ramp (8 hits per code) gives |DNL|max 0.16 and |INL|max 0.11 LSB, and the gain error is +2.37 %. Total power is 91 + 144 uW.

The extracted layout draws more power, mostly from VDDD. Extraction adds wiring capacitance to every logic net, and the two clock nets that toggle every cycle (the clock buffer output, 57 fF, and the comparator clock, 35 fF) alone account for about 30 uW at 100 MHz and 1.8 V. The extracted ramp shows no code toggling and none of the schematic's carry-transition errors, so its DNL/INL are lower than the schematic's. Because the cause of the schematic error is not isolated (Section 3), the schematic figures should be treated as the design's linearity until it is.

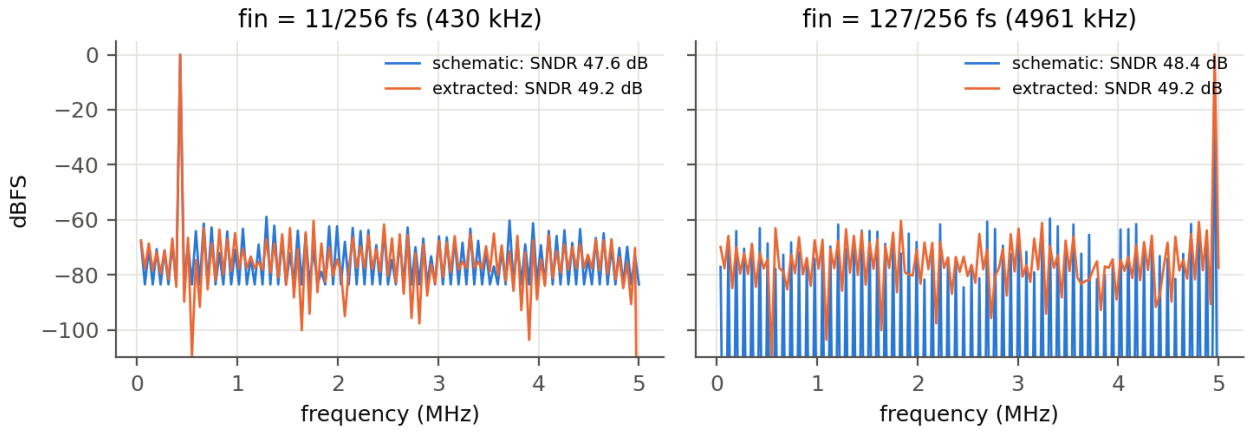


Figure 16. Output spectra, schematic and extracted layout, at 430 kHz and 4.96 MHz.

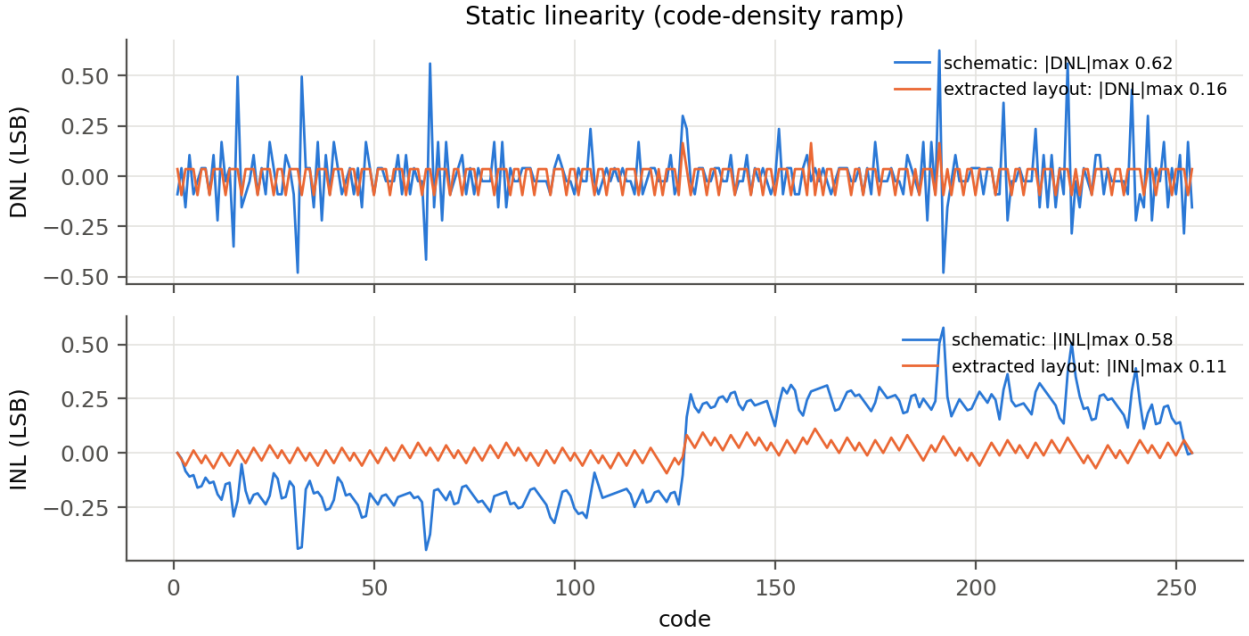


Figure 17. DNL and INL from the code-density ramp, schematic and extracted layout.

7. Limitations

• Not fabricated. All results are simulations; nothing was characterised in silicon.

• Supply, reference and clock are ideal sources. VREF = VDDA, and there is no reference buffer, decoupling or bond-wire model. A real reference would need to supply the DAC's switching charge in under 5 ns.

• Extraction is capacitance only. Wire resistance (mainly the long met3 top-plate and driver routes) is not modelled.

• Capacitor mismatch is simulated with the behavioural model, because the PDK's SPICE mismatch libraries do not randomise cap_mim. Transistor mismatch is simulated at transistor level.

• PVT corners use the schematic netlist. The extracted netlist was simulated only at TT, 27 C.

• There are no metal fill, density, antenna or ESD/IO checks, and no pad ring. The block is a macro, not a chip.

• Dynamic results use 256-point FFTs and the extracted-layout ramp uses 8 hits per code (schematic 15), which limits the resolution of SFDR and DNL.

8. Tool flow

Step	Tool
Schematic capture / netlisting	xschem 3.4.8RC (schematics written by a Python generator)
Simulation	ngspice 47, sky130A models (open_pdks 0fe599b)
Layout	Magic 8.3 (Tcl, script-generated) + sky130_fd_sc_hd cells
DRC	Magic sky130A deck; KLayout 0.30 sky130A_mr deck
LVS	netgen 1.5
Extraction	Magic ext2spice (C + coupling C)
Analysis / figures	Python 3 (numpy, scipy, matplotlib); KLayout and xschem renders

All figures were generated by the tools listed above (simulation data, KLayout renders of the GDS, xschem schematic exports). Every number in this report is read from the project's params.json, which the stage scripts write.